Janavahini: how the system is built and how an alert flows

Counters on every road, gate and junction estimate crowd density, look a minute ahead, and pass warnings back along the pilgrim route to the police who can act. Decisions are made on the devices, so alerts keep working when the network or the command centre fails.

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Challenge RTIH × RMC Innovation-Led Solutions for Godavari Pushkaralu 2027 **Date** November 2026

1 · TECHNICAL ARCHITECTURE

Four layers: counting, node, mesh, server, with the RMC ICCC on top

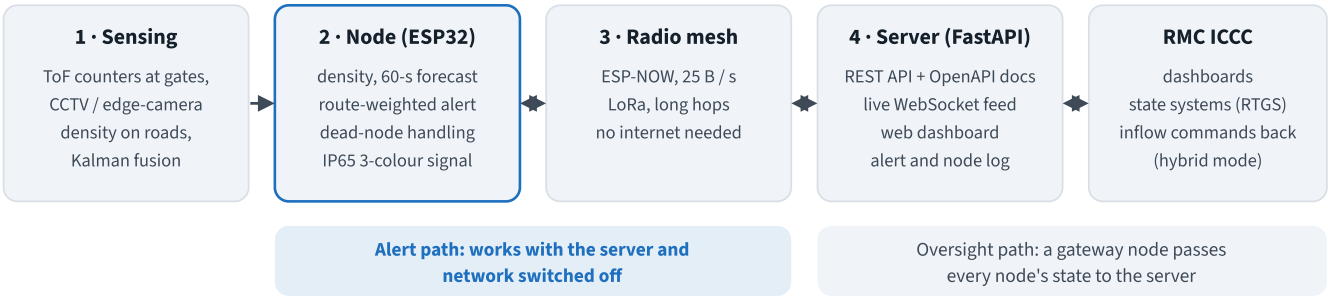


Figure 1. System architecture. Each node applies the stronger of its own decision and the ICCC's command, so the ICCC can only add caution.

Layer	Technology	Prototype status
1 · Sensing	A pair of VL53L1X time-of-flight sensors per 1.2 m lane gives direction-aware in/out counts at gates (4 lanes) and exits (2–3 lanes), where barricades already form lanes (shaded: direct sun cuts their range to under 1 m). On wide roads, density from existing CCTV or an edge camera wired to the node. A 2-state Kalman filter per node, [density, counter drift], fuses them so counter errors cannot pile up; on a patch without counters, its allowed rate of change is set by physics (peak walking flow × boundary width ÷ area), so it follows a fast-filling patch within seconds.	Modelled with realistic errors and tested at site scale (E12) and in the JuPedSim compartment (E16); ported to the firmware (matches within 0.001 persons/m^2)
2 · Node	ESP32 firmware (C++, PlatformIO) computes density, a 60-s forecast and the route-weighted alert, and drives an IP65 three-colour signal. The alert logic is one header shared with the simulator.	Complete; 22% RAM; 244 virtual boards tested; not yet flashed
3 · Mesh	ESP-NOW 25-byte messages once a second for short hops (0.7% channel load with 10 boards in range); LoRa for long hops, sent on change. Silent nodes are detected and reported.	ESP-NOW done; LoRa stubbed until modules arrive
4 · Server	Python FastAPI: REST with OpenAPI documentation, WebSocket live feed; web dashboard with zone map, alert log, inflow commands and node-down reports	Done; end-to-end tests pass

2 · WORKFLOW INSIDE EVERY NODE · REPEATED ONCE A SECOND

Count, look ahead, combine warnings, signal, pass back

1 · Measure
Line counters at gates and exits count people in and out; on wide roads a camera density reading arrives every 10–120 s.

2 · Fuse
Kalman filter: predict density from (in − out) ÷ area, correct with each density reading; it also learns the counter drift.

3 · Look ahead
Forecast = density now + rate of rise × 60 s (rate used only once the patch is busy, ≥ 1.5 persons/m^2).

4 · Listen
Receive the states of downstream nodes over the mesh; weight each by the share of this road's flow that heads there (from the route map).

5 · Combine
Take the strongest of: own forecast, weighted downstream warnings, and the ICCC command (if the network is up).

6 · Dead neighbour?
"Unknown is not safe": keep its last warning, listen one hop further, report "NODE DOWN – send staff".

7 · Signal
Set the three-colour light: green, amber or red (NDMA-aligned levels).

8 · Pass back
Broadcast state to upstream neighbours and, via the gateway, to the server and ICCC.

3 · OPERATIONAL WORKFLOW · WHO DOES WHAT DURING A SURGE

From the first sign of filling to a safe release

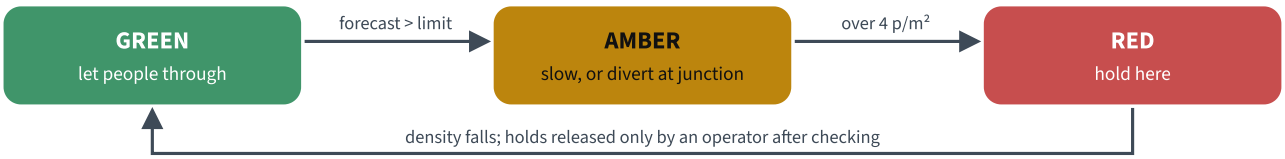


Figure 2. Alert states seen by the police at each post. Roads that don't feed the problem stay green.

Step	Counter node (automatic)	Police post / volunteers	ICCC operator
1 · Normal	Counts and forecasts every second; light green	Let people through	Watches live density map
2 · Early warning	Ghat forecast to cross 4 p/m^2 → amber passed back along the route	Road head: slow the flow. Junction: send new arrivals to the ghat with room	Sees which roads feed the problem; may add caution
3 · Danger	Ghat over the limit → red passed back	Hold at the nearest upstream holding area, where people can wait safely	Coordinates diversions, PA and police reinforcements
4 · Node failure	Neighbours keep its warning; "NODE DOWN – send staff"	Staff check the node	Clears the report after the check
5 · Release	Density falls; forecast back below limit	Release in stages	Confirms release of holds

4 · HARDWARE, SOFTWARE, INFRASTRUCTURE AND INTEGRATION

Item	Requirement
Hardware per node	ESP32 board with LoRa (SX1276); IP65 three-colour signal tower; IP65 enclosure; mains adapter with backup battery. Gate and exit nodes: a pair of VL53L1X ToF sensors per lane (8 at a gate, 4–6 at an exit), shaded. Wide-road nodes: a density feed from existing CCTV or an edge camera box, wired to the node.
Quantity	244 nodes for the full site (136 gate and exit counter nodes; 108 density nodes at compartments, junctions, roads and Nagaras) plus 10 zone gateways bridging the mesh to the ICCC over 4G; 11 nodes and a gateway for the pilot.
Software	Open-source ESP32 firmware, FastAPI backend and web dashboard; runs on one server or VM at the ICCC.
Infrastructure	Power and mounting points at gates, roads and junctions. Counters only where barricades already form lanes; no road is narrowed for counting.
Integration	(a) Read crowd counts from existing CCTV/AI analytics, at the node where possible so a network cut does not blind it. (b) Push node states, alerts and node-down events to ICCC dashboards and RTGS via a documented REST/WebSocket API. (c) Accept ICCC inflow commands. (d) Staff messages in Telugu and English.

5 · DATA AND SECURITY

- **Data:** only counts and densities per patch, node health and alert logs. Cameras count on the device: no images leave it, no video is stored, no faces, phones or personal data.
- **Sensor requirement (simulation, E12):** a density reading at least every 2 minutes with at most 20% error per reading. The design stayed safe with half the readings lost, with a 20% camera undercount, and when any single camera or board failed.
- **Security:** authenticated node messages so a spoofed radio cannot raise or clear alerts; TLS on all API traffic; API keys and role-based access for ICCC operators; an audit log of every command; holds released only by an operator; hosting in the RMC or state data centre following CERT-In guidelines.

6 · PROTOTYPE IN ACTION

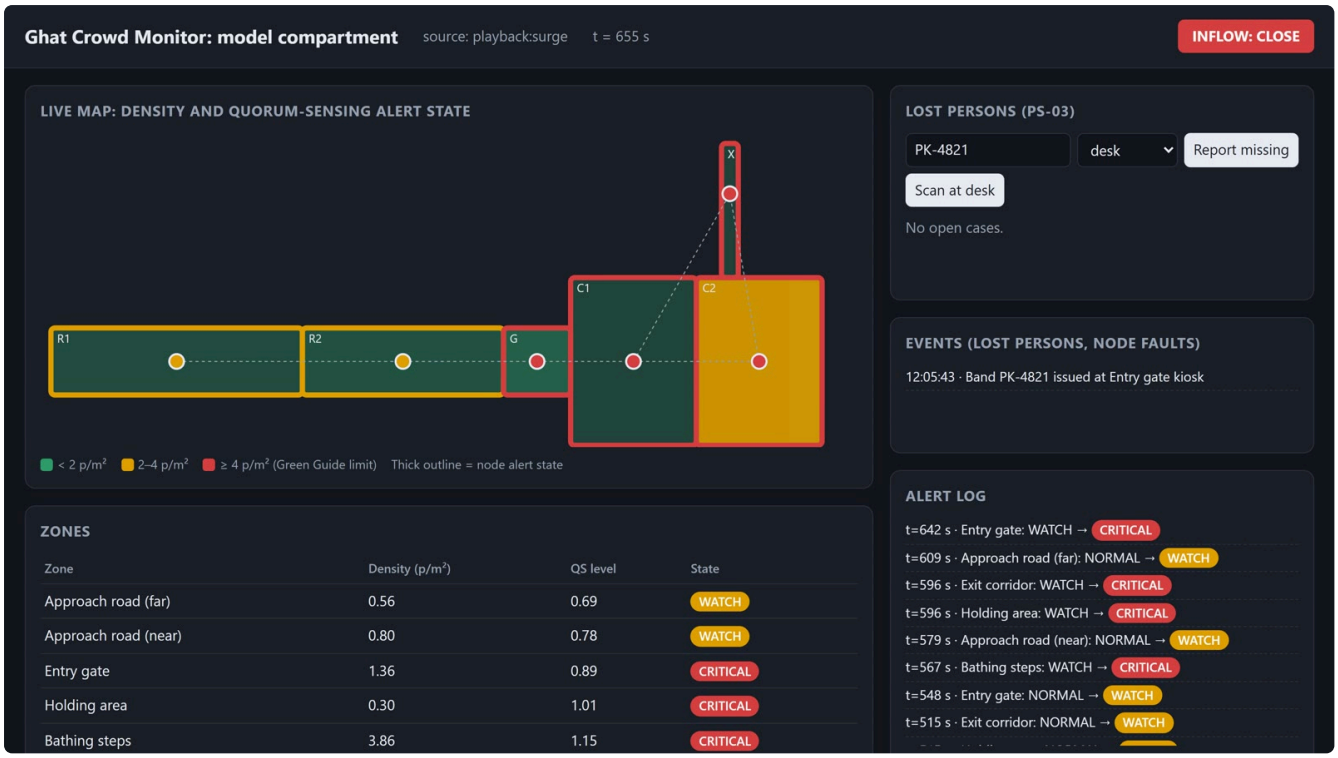


Figure 3. Our prototype dashboard replaying a simulated surge. The warning has reached the approach road (amber) before the bathing steps pass the limit. All results so far are from simulation; the board firmware is not yet field-tested.